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When Does External Guidance Help LLM Reasoning?

A Bias-Variance Theory of Guidance-Augmented GRPO Derives a Closed-Form Optimal Guidance Weight and Proves an Unavoidable δ^2 Bias Floor

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Adding teacher demonstrations or search guidance to GRPO can help or hurt LLM reasoning depending on how far the guidance distribution is from the current policy. GA-GRPO bounds this with a total-variation divergence term, derives a closed-form MSE-optimal guidance weight, and proves a minimax lower bound showing the resulting $\Omega(\delta^2)$ bias is unavoidable. Using the optimal weight, Qwen2.5-Math-7B-Base gains 2.3% on MATH-500 and uses 31% fewer GPU-hours than vanilla GRPO.

AT A GLANCE

Problem: Guidance-augmented GRPO methods tune guidance weight by grid search; no principled theory exists for when guidance helps.

Why it matters: Optimal guidance weight determines whether teacher signals accelerate or degrade RL training for LLMs.

Method: Model guidance as a sampling-distribution shift; derive closed-form α^* from the bias-variance MSE formula.

Result: GA-GRPO with α^* achieves 82.4% on MATH-500 vs. 80.1% for vanilla GRPO, using 31% fewer GPU-hours.

Evidence: Nine math benchmarks; Qwen2.5-Math-7B-Base vs. TAPO, LUFFY, ExPO, and vanilla GRPO.

The Big Picture

Reinforcement learning from verifiable rewards (RLVR) is now the dominant post-training strategy for reasoning LLMs. Recently, methods such as TAPO, LUFFY, and ExPO add external guidance—teacher CoT demonstrations, search results— to the GRPO rollout distribution and report empirical gains. But *when* does guidance help, and how much should be used?

Torres et al. provide the first rigorous answer: a bias-variance decomposition of guidance-augmented GRPO that produces a closed-form optimal guidance weight and a matching minimax lower bound showing the resulting bias is theoretically unavoidable.

Why Existing Approaches Fall Short

Vanilla GRPO / REINFORCE: Purely on-policy; zero bias but high variance for sparse rewards, requiring many samples.

SFT from demonstrations: High bias (model drifts toward teacher, away from its own distribution), no principled debiasing.

Ad-hoc guidance weighting (TAPO, LUFFY, ExPO): Empirically effective but guidance weights (ranging 0.01–10 across papers) are chosen by grid search with no convergence guarantee.

GA-GRPO: Derives α^* from first principles and converges at $\mathcal{O}(1/\sqrt{T})$ to a neighborhood whose radius is bounded by guidance bias.

Core Mechanism: Mixture Sampling and Importance Weighting

Guidance-Augmented GRPO draws rollouts from:

$$\tilde{q}_t = (1-\alpha)\, p_{\theta_t} + \alpha\, q$$

where q is the guidance distribution and $\alpha \in [0,1]$.

The gradient estimator uses importance weights to correct for off-policy sampling:

$$\hat{g}_t = \frac{1}{K}\sum_{k=1}^{K} \frac{p_{\theta_t}(\tau_k)}{\tilde{q}_t(\tau_k)}\, r(\tau_k)\, \nabla_\theta \log p_\theta(\tau_k)$$

The MSE of $\hat{g}_t$ decomposes as:

$$\mathrm{MSE}(\alpha) = \frac{\sigma^2(\theta_t)}{K(1-\alpha)} + \alpha^2 C^2 \mathrm{TV}^2(q, p_{\theta_t})$$

The first term is variance (decreases with K and α); the second is squared bias (grows with α).

Technical Formulation

Minimising $\mathrm{MSE}(\alpha)$ over $\alpha \in [0,1]$ yields the closed-form optimal guidance weight:

$$\alpha^* = \frac{\sigma^2(\theta_t)/K}{\sigma^2(\theta_t)/K + 2C^2\, \mathrm{TV}^2(q, p_{\theta_t})}$$

This is small when $\mathrm{TV}(q, p_\theta)$ is large (biased teacher) and grows as the model improves toward the teacher.

Convergence theorem (informal). Under L-smooth policy gradients and $\mathrm{TV}(q, p_\theta) \le \delta$, GA-GRPO with step size $\eta_t = \mathcal{O}(1/\sqrt{t})$ satisfies:

$$\frac{1}{T}\sum_{t=1}^{T} \|\nabla \mathcal{L}(\theta_t)\|^2 \le \frac{C_1}{\sqrt{T}} + \delta^2 C_2$$

Minimax lower bound. For any guidance-augmented algorithm with $\mathrm{TV}(q, p_\theta) \geq \delta$:

$$\frac{1}{T}\sum_{t=1}^{T} \|\nabla \mathcal{L}(\theta_t)\|^2 \geq \Omega\left(\frac{1}{\sqrt{T}}\right) + \Omega(\delta^2)$$

The $\Omega(\delta^2)$ bias floor is *unavoidable*—no algorithm using guidance with fixed divergence δ can eliminate it.

Learning or Inference Procedure

1. Specify teacher q; estimate $\mathrm{TV}(q, p_\theta)$ via Pinsker's inequality on a calibration set.
2. Compute α^* using the closed-form; update every 100 gradient steps.
3. Run GA-GRPO with mixture $\tilde{q}_t$.
4. Evaluate on held-out math benchmarks.

What the Guarantee Says

GA-GRPO converges at $\mathcal{O}(1/\sqrt{T})$ with a bias floor of $\delta^2 C_2$. The optimal α^* minimises MSE at each step, making it the best possible guidance weight under the assumed smoothness and boundedness conditions. The minimax lower bound confirms no improvement on δ^2 is achievable by any algorithm.

Experimental Findings

- **Model:** Qwen2.5-Math-7B-Base; teacher: GPT-4o CoT (50K examples).
- **Benchmarks:** MATH-500, AIME 2024/25, AMC 2023, GPQA-Diamond, OlympiadBench, Minerva, Olympus, GaoKao (9 total).
- GA-GRPO (α^*): **82.4%** on MATH-500 vs. 80.1% vanilla GRPO.
- Surpasses TAPO (+1.1%), LUFFY (+0.7%), ExPO (+2.3%) averaged across benchmarks.
- **31% fewer GPU-hours** (3.2M vs. 4.6M GPU-minutes) at matched final accuracy.

Ablations and Interpretation

Fixed α vs. α^*: Fixed α=0.3 achieves 80.8%; α^* achieves 82.4%; the adaptive scheduling is the key improvement.

TV estimation: KL-based estimation is within 0.5% of exact TV; entropy-based proxies are noisier but still achieve 80.2%.

Low-quality teacher (δ=0.8): α^* drives toward zero, recovering vanilla GRPO—confirming the method is teacher-agnostic.

δ^2 bias empirical confirmation: High-divergence teacher leads to lower accuracy despite large guidance signal, matching the lower bound.

Reference

Sofia Torres, Gabriel Almeida, Carter Adams, Camila Rocha. **When Does External Guidance Help LLM Reasoning? A Bias-Variance Theory of Guidance-Augmented GRPO**. arXiv:2610.06861, October 2026. https://arxiv.org/abs/2610.06861